

1. Display employees full name and department name for all employees

```
inventory> db.employee.aggregate([
...   {
...     $lookup: {
...       from: "department",
...       localField: "department_id",
...       foreignField: "department_id",
...       as: "departments",
...     },
...   },
...   {
...     $project: {
...       _id: 0,
...       full_name: 1,
...       "departments.department_description": 1
...     }
...   }
... ]);
[
  {
    full_name: 'Sheri Nowmer',
    departments: [ { department_description: 'HQ General Management' } ]
  },
  {
    full_name: 'Derrick Whelply',
    departments: [ { department_description: 'HQ General Management' } ]
  },
  {
    full_name: 'Michael Spence',
    departments: [ { department_description: 'HQ General Management' } ]
  },
  {
    full_name: 'Maya Gutierrez',
    departments: [ { department_description: 'HQ General Management' } ]
  },
  {
    full_name: 'Roberta Damstra',
    departments: [ { department_description: 'HQ Information Systems' } ]
  }
]
```

2. Display employees with position “VP Country Manager” (only display employee full name and salary).

```
inventory> db.employee.find({position_title:"VP Country Manager"},{full_name:1,salary:1,_id:0})
[
  { full_name: 'Derrick Whelply', salary: 40000 },
  { full_name: 'Michael Spence', salary: 40000 },
  { full_name: 'Maya Gutierrez', salary: 35000 },
  { full_name: 'Beverly Baker', salary: 30000 },
  { full_name: 'Pedro Castillo', salary: 35000 },
  { full_name: 'Laurie Borges', salary: 35000 }
]
```

3. Display customers full names and their regions.

```
inventory> db.customer.aggregate([
...   {
...     $lookup: {
...       from: "region",
...       localField: "address.customer_region_id",
...       foreignField: "region_id",
...       as: "customers_regions"
...     }
...   },
...   {
...     $project: {
...       _id: 0,
...       full_name: 1,
...       "customers_regions.sales_region": 1
...     }
...   }
... ])

[
  { customers_regions: [ { sales_region: 'Mexico South' } ] },
  { customers_regions: [ { sales_region: 'Canada West' } ] },
  { customers_regions: [ { sales_region: 'North West' } ] },
  { customers_regions: [ { sales_region: 'Canada West' } ] },
  { customers_regions: [ { sales_region: 'Central West' } ] },
  { customers_regions: [ { sales_region: 'North West' } ] },
  { customers_regions: [ { sales_region: 'Mexico South' } ] },
  { customers_regions: [ { sales_region: 'Mexico Central' } ] },
  { customers_regions: [ { sales_region: 'Canada West' } ] },

```

4. In product find all products that was branded by " Washington "

```
inventory> db.product.find({brand_name: "Washington"},{})
[
  {
    _id: ObjectId('53c7c973cccf26e6de850bd88'),
    product_class_id: Long('30'),
    product_id: Long('1'),
    brand_name: 'Washington',
    product_name: 'Washington Berry Juice',
    SKU: Long('90748583674'),
    SRP: 2.85,
    gross_weight: 8.39,
    net_weight: 6.39,
    recyclable_package: false,
    low_fat: false,
    units_per_case: Long('30'),
    cases_per_pallet: Long('14'),
    shelf_width: 16.9,
    shelf_height: 12.6,
    shelf_depth: 7.4
  },
  {
    _id: ObjectId('53c7c973cccf26e6de850bd89'),
    product_class_id: Long('52'),
    product_id: Long('2'),

```

5. Group products by brand name, count number and display the result as follow

```
inventory> db.product.aggregate([
...   {
...     $group: {
...       id: "$brand_name",
...       TotalCount: { $sum: 1 }
...     }
...   }
... ])
[
  { _id: 'Jardon', totalCount: 5 },
  { _id: 'Best Choice', totalCount: 36 },
  { _id: 'Blue Label', totalCount: 22 },
  { _id: 'Washington', totalCount: 11 },
  { _id: 'Footnote', totalCount: 2 },
  { _id: 'Steady', totalCount: 19 },
  { _id: 'Quick', totalCount: 2 },
  { _id: 'Club', totalCount: 20 },
  { _id: 'Cormorant', totalCount: 32 },
  { _id: 'Symphony', totalCount: 1 },
  { _id: 'Gauss', totalCount: 5 },
  { _id: 'Best', totalCount: 4 },
  { _id: 'Token', totalCount: 11 },
  { _id: 'Radius', totalCount: 4 },
  { _id: 'Excel', totalCount: 5 },
  { _id: 'Monarch', totalCount: 5 },
  { _id: 'Special', totalCount: 4 },
  { _id: 'Jumbo', totalCount: 5 },
  { _id: 'Tri-State', totalCount: 44 },
  { _id: 'Plato', totalCount: 29 }
]
```

6. Group products by brand_name and product_name ,only select brand names ("Blue Label","King","Washington") then sort them by brand_name and product_name ascending and display result as follow

```
inventory> db.product.aggregate([
...   {
...     $match: {
...       brand_name: { $in: ["Blue Label", "King", "Washington"] }
...     }
...   },
...   {
...     $group: {
...       _id: {
...         brand_name: "$brand_name",
...         product_name: "$product_name"
...       }
...     }
...   },
...   {
...     $sort: {
...       "_id.brand_name": 1,
...       "_id.product_name": 1
...     }
...   }
... ])
[
  {
    _id: { brand_name: 'Blue Label', product_name: 'Blue Label Beef Soup' }
  },
  {
    _id: { brand_name: 'Blue Label', product_name: 'Blue Label Canned Beets' }
  },
  {
    _id: { brand_name: 'Blue Label', product_name: 'Blue Label Canned Peas' }
  },
  {
    _id: {
      brand_name: 'Blue Label',
      product_name: 'Blue Label Canned String Beans'
    }
  },
  {
    _id: { brand_name: 'Blue Label', product_name: 'Blue Label Canned Tomatos' }
  },
]
```